Factory-to-Customer Shipping Route Efficiency Analysis (Nassau Candy Distributor)

This is a data analytics project I built to practice the full pipeline a data analyst would actually use on the job: cleaning messy source data in Python, answering business questions in SQL, and presenting the results in Power BI.

Dataset: 10,194 order-level shipment records, 18 fields, covering the US and Canada, January 2024 through December 2025.

The business problem

Nassau Candy Distributor ships candy from a handful of factories out to customers across North America. They have plenty of order and shipment data sitting around, but nobody had ever turned it into route-level intelligence, meaning there was no clear answer to which shipping lanes were actually working well and which ones weren't. Decisions about carriers and shipping methods were being made on gut feel rather than evidence.

Questions I set out to answer:

- Which factory-to-customer routes are consistently fast, and which are chronically slow?
- Which states or regions act as bottlenecks (lots of volume, but also slow)?
- How do the different shipping methods (Standard, Second Class, First Class, Same Day) actually trade off speed against cost?
- Where should logistics investment go first?

My approach: clean and validate the raw data in Python, build out route-level and lead-time features, answer the business questions in SQL, then put the findings in front of a non-technical audience through an interactive Power BI dashboard.

Dataset description

Field	Type	What it is
Row ID	Integer	row identifier
Order ID	String	order identifier
Order Date	Date (DD-MM-YYYY)	when the order was placed
Ship Date	Date (DD-MM-YYYY)	when it shipped

Field	Type	What it is
Ship Mode	Categorical	Standard Class / Second Class / First Class / Same Day
Customer ID	Integer	customer identifier
Country/Region	Categorical	United States / Canada
City	String	customer city
State/Province	String	customer state or province, 59 unique values
Postal Code	String	customer postal or ZIP code
Division	Categorical	Chocolate / Sugar / Other
Region	Categorical	Pacific, Atlantic, Interior, Gulf
Product ID	String	product identifier
Product Name	String	e.g. "Wonka Bar - Milk Chocolate"

Field	Type	What it is
Sales	Float	total sales value on the order line
Units	Integer	units ordered
Gross Profit	Float	Sales minus Cost
Cost	Float	cost to manufacture

There's also a small reference table mapping each product to the factory that makes it (five factories total: Lot's O' Nuts, Wicked Choccy's, Sugar Shack, Secret Factory, The Other Factory). That mapping is what lets an order-level dataset get turned into a route-level one, since "route" here means factory of origin to customer state.

A data quality problem worth mentioning up front

The raw Ship Date column is not usable row by row. I checked the gap between Order Date and Ship Date across all 10,194 rows and every single one of them was off by somewhere between 904 and 1,642 days, with only 28 distinct values in the whole column. That's not normal messy data, that's a systematically

broken field. Deleting the bad rows wasn't an option since there were no good ones left.

So instead I derived a lead time estimate from Ship Mode instead (Standard Class, Second Class, First Class, Same Day each get an estimated typical duration). I'm flagging this clearly rather than pretending it's measured data, because any chart built on top of it (state comparisons, route rankings, etc.) is really reflecting each area's mix of shipping methods rather than a true measured transit time. Worth knowing before drawing conclusions from it.

Key findings

- Standard Class carries 60% of total sales ($85,490 of $141,784) at an estimated 4.97 day lead time. Same Day cuts that down to 0.52 days but only carries 5% of sales ($7,114). The faster tiers are underused relative to what they'd actually buy in speed. Worth testing something like an automatic upgrade rule for orders above a certain value.
- Chocolate makes up almost the entire business, 92.9% of total sales ($131,693 of $141,784). Sugar and Other combined are only 7.1%. Worth asking whether those smaller divisions need their own shipping lanes at all, or whether they should just ride along on Chocolate's existing routes.

- Route speed varies more than you'd expect. The fastest routes that had enough volume to be meaningful (Secret Factory to Colorado, Secret Factory to Virginia) average around 2.8 to 3.0 days. The slowest cluster (Lot's O' Nuts to Maine, Vermont, and Newfoundland & Labrador) sits at 4.97 days, nearly double. That gap is where I'd start if I were prioritizing which carrier contracts to renegotiate.

Project structure

```
nassau-candy-shipping-analysis/
  README.md
  data/
    Nassau_Candy_Distributor.csv
(raw source)
    Nassau_Candy_Distributor_clean.csv
(output of data_cleaning.py)
  data_cleaning.py
  queries.sql
  powerbi_dashboard_spec.md
  schema.md
  powerbi/
    Shipping_Route_Dashboard.pbix
```

How to run it

Python cleaning step:

```
pip install pandas
python data_cleaning.py
```

This reads the raw CSV, checks and standardizes it, builds the lead time and route fields, and writes out the cleaned CSV.

SQL step: load the cleaned CSV into whatever database you're using (MySQL, Postgres, SQLite, anything with standard SQL support) as a table called `shipments`, then run through `queries.sql` in order. Each query has a comment above it explaining what business question it's answering.

Power BI step: open Power BI Desktop, import the cleaned CSV, and build the dashboard following `powerbi_dashboard_spec.md`.

Data model / schema

See `schema.md` for the full schema, both the flat table this project actually uses and the star schema version I'd build if this were going into production at a larger scale.

Tech stack

Python (pandas), SQL, Power BI (DAX), and Plotly.js for the browser demo version.

Notes

This was built as a self-directed portfolio project simulating a logistics analytics engagement. The numbers throughout are computed from the actual cleaned dataset. The Ship Mode based lead time is a documented workaround for a real data quality problem in the source file, not something to present as measured fact without the caveat attached.